Validating HTML

The Five Rules for Universal Code

web explorations

by Peter K. Johnson

History of HTML copy

Tim Berners-Lee borrowed from an established mark-up language SGML (Standardized Generalized Markup Language) first developed by IBM in the 1960's in an attempt to make government documents readable by computers.

Over the next 10 years several browsers were developed including Internet Explorer and Netscape. Each new version introduced competing features and it wasn't long before web developers had to write two different versions of every web page.

Tim Berners-Lee organized the W3C (World Wide Web Consortium) bringing all the major players in the industry together in order to create HTML standards so the same markup code would work the same in any browser.

At the same time the W3C created a new standard, XML (Extensible Markup Language) that allowed developers to create their own tags. XML is used throughout the industry allowing various computer systems to exchange data. Phone records, graphic images, real estate listings are all examples of XML data. In order for XML to work however, developers needed to follow specific rules.

Meanwhile, back in the HTML circles, developers were creating web pages with all types of markup. The browsers were written to display pages even though tags might be missing or put in the wrong order. It was all left up to the browsers. Web developers were just excited to be able to create a web page. "We don't need no stinkin' rules…." they said.

But, browsers got more involved and new browsers were being developed. Mobile phones were being introduced and memory suddenly was at a premium. There was no room for the huge complicated browsers that everyone was using on their desktop machines.

As XML became more popular a new version of HTML called XHTML was developed combining both HTML and using the XML rules. This standard was created in parallel to

HTML and has since been dropped by the W3C standards group. (Even though the standards are no longer supported, the web pages still work in all the current browsers.)

Meanwhile the HTML standards group continued working on newer versions. The current version as of July 2013 is HTML5.

Anatomy of HTML

HTML is made up of several different pieces.

TAGS

Start and stop tags are used to tell the browser when something starts and when it ends.

For example <p> starts a paragraph and </p> ends the same paragraph.

ELEMENTS

Everything inside a set of tags, including the tags themselves is called an element.

In the graphic below the <head> <title> and <body> elements are all marked in green.

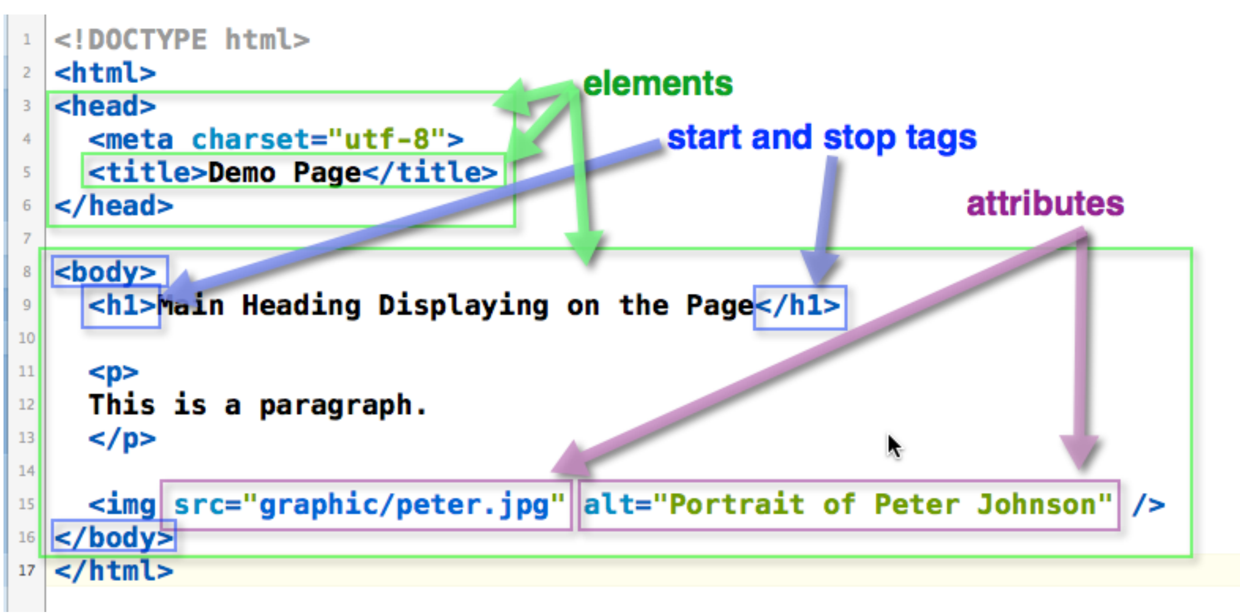
Each set of start/stop tags, and everything enclosed by them, make up an element.

ATTRIBUTES

Sometimes you want to give the browser more information. Attributes are used to do this.

An attribute in HTML is like an adjective in English.

An example of attributes are the src=" " and alt=" " attributes that are added to every <img> element.



DOCTYPE and charset

The DOCTYPE is used to tell the browser what version of HTML to use.
The W3C has greatly simplified this markup element for HTML5. All you have to do is add "html".
This should be the first line on every web page you create:
<!DOCTYPE html>

The Web has unified the globe. But that also means that a method had to be developed so the browser would know which language to use. Unicode was developed giving every character in every written language a number so it can be represented on a web page (and by other computer programs.)

Here is the Unicode website with charts for each language: http://www.unicode.org/charts/

The most universal character set is UTF-8. It can represent every character in the Unicode library and has become the default character encoding for most operating systems, programming languages, and browsers around the world.

You should always include this <meta> element in the <head> element of every web page:
<meta charset="utf-8" />

```html
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>Demo Page</title>
</head>

<body>
  <h1>Main Heading Displaying on the Page</h1>

  <p>
  This is a paragraph.
  </p>

  <img src="graphic/peter.jpg" alt="Portrait of Peter Johnson" />
</body>
</html>
```

The DOCTYPE tells the browser what version HTML to use.

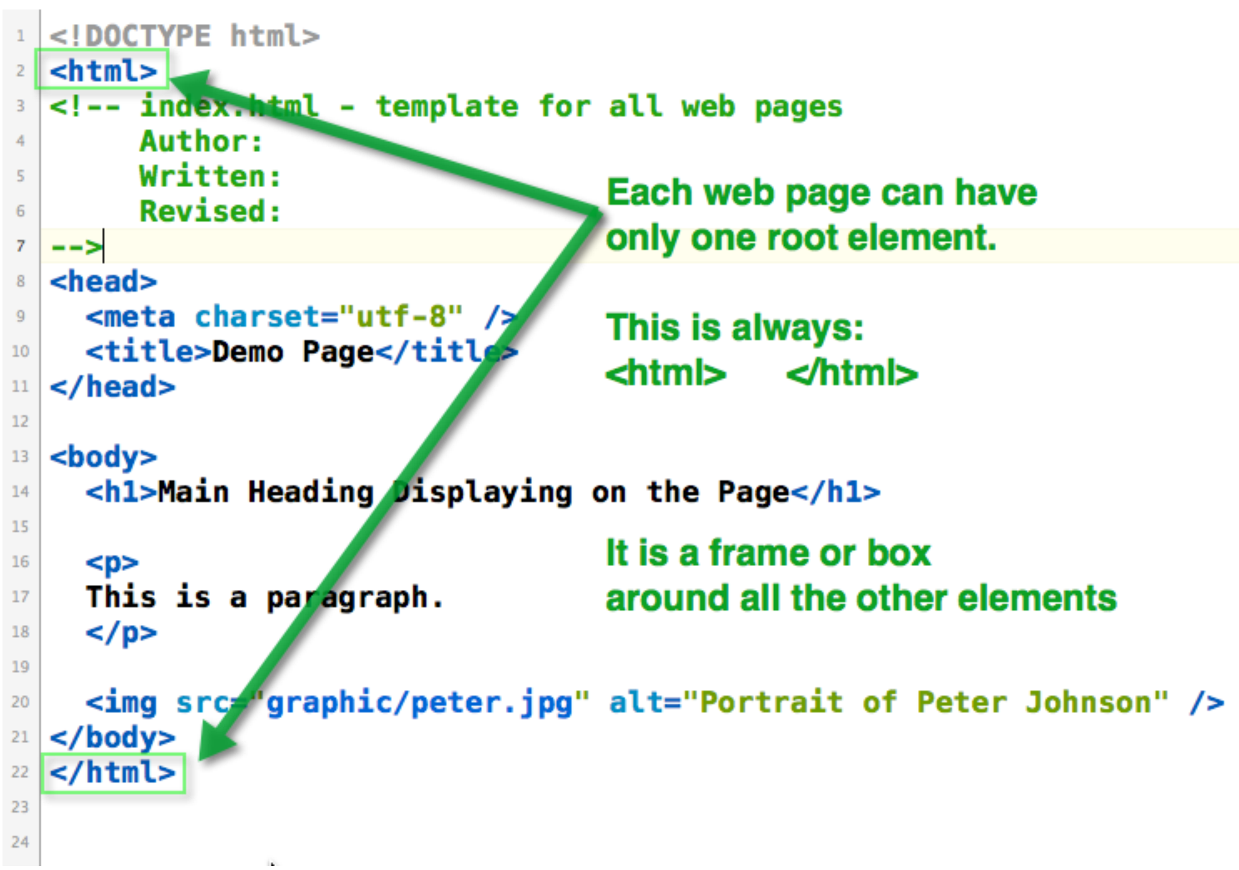
The charset tells the browser which alphabet to use.

Rule 1 - Only One Root Element

Each web page can have only one root element.

For web pages this is always <html> and </html>. All the other elements on the page must go inside of these two tags.

Anything that comes after the </html> may be ignored. This is especially true for mobile devices. Due to their limited memory the browsers on mobile devices don't have enough room in their program code to check for exceptions such as text outside of the <html> element. If content if put after the </html> it is simply dropped.



Rule 2 - Tags are Case Sensitive

Tags must match and they are case sensitive.

Based on this rule you can safely write all your tags and attributes using lower case.

Technically you could mix and match UPPER/lower case, but it has become standard to write all tags in lower case.

Rule 3 - Attribute values must have quotes

Attribute values must be surrounded by quotes.

An attribute is like an adjective to a tag. It gives more details to the browser describing how that element should be displayed.

Each element has its own specific set of attributes. These are part of the standard that was created by the W3C. You can look up a specific element on W3Schools.com to see what attributes are available for that element.

All attributes must be inside of the actual element itself or in between the < and > of the element.

The <img /> element is a good example. The two attributes src=" " (name of the source file or image) and alt=" " (alternative text if the source file is not available) tell the browser more about the <img /> element.

Every attribute has an equal sign "=" as well as a value that must be enclosed using quotes.

```
<img src="graphic/peter.jpg" alt="Portrait of Peter Johnson" />
```

Two attributes for the <img /> element

Rule 4 - Each Start tag needs a matching Stop Tags

For every action there is a reaction.

For every on there is an off. For every start a stop. These are the laws of science as well as of browsers.

If you have a tag you need to have a matching (pronounced "stop strong", the / means stop.)

This works great for tags that are paired. But, what about the single tag elements such as:

- <img /> - image display
- <meta /> - meta tag
- <hr /> - horizontal rule
-
 - line break

These have the stop tag built in.

Notice the space before the stop slash "/"?

The space is used to fool old browsers. When a browser reads a tag that isn't in it's dictionary, it simply discards it. By including a space before the stop slash "/" the older browsers see and recognize the <br and display a line break.

Without the space the old browsers would see
 which isn't in their dictionary and it would be discarded without a line break being displayed.

Don't confuse this with all the other elements that have double tags.
There's a difference in the order.

Double tag elements have the stop slash "/" in the front, right after the < in the closing tag:

</p>

Single tag elements have the slash right before the > at the end:

Rule 5 - Elements must be properly nested.

All elements must be properly nested. That means that if you start with a strong tag and then a paragraph, you have to have the stop paragraph first and then the strong tag.

For example:

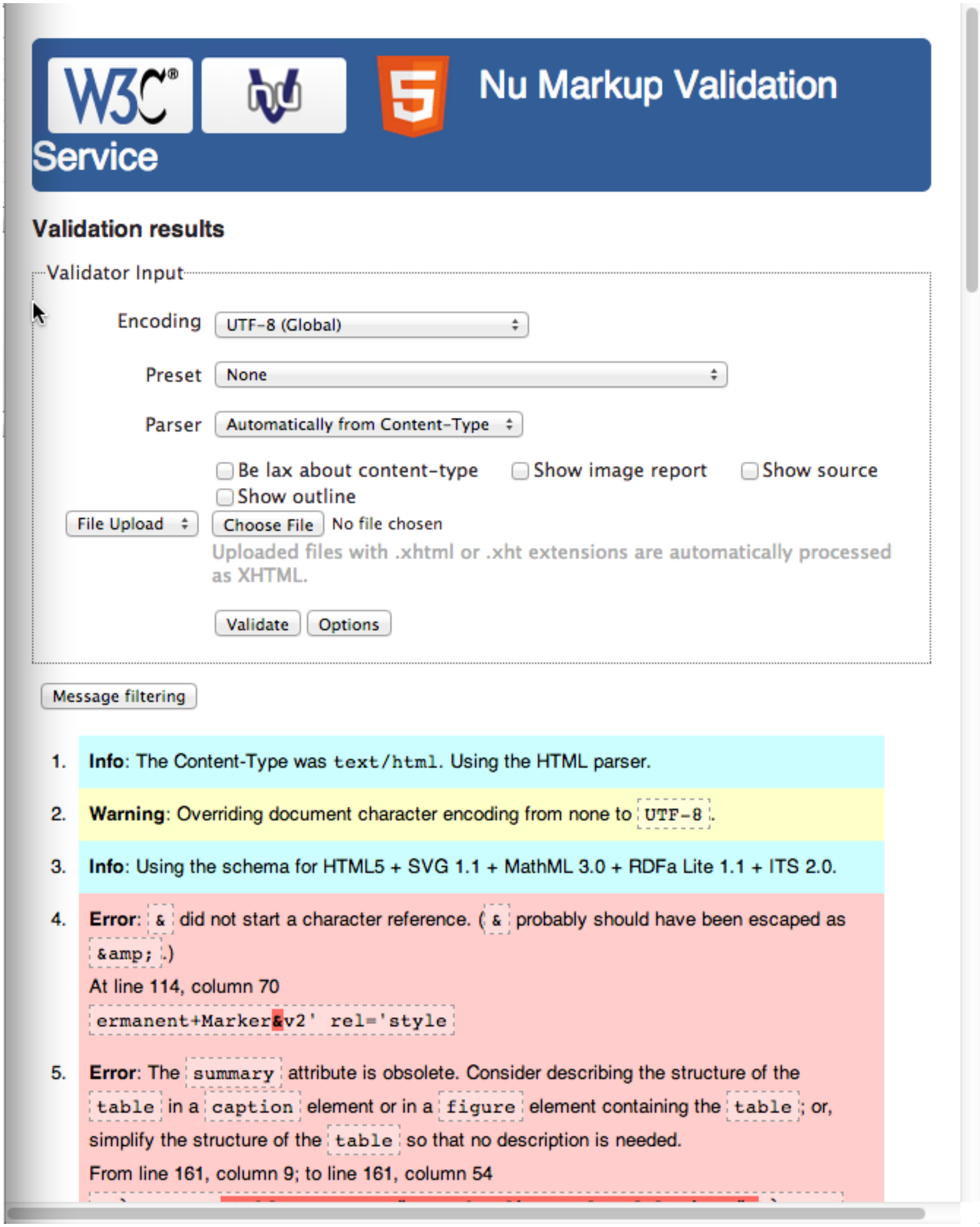
This is correct: <p>This is a paragraph.</p>
This is not valid: <p>this is a paragraph.</p>

Validation Tools

As you write your HTML code you need to keep these five rules in mind so your code will be valid and your pages will display no matter what browser someone else uses.

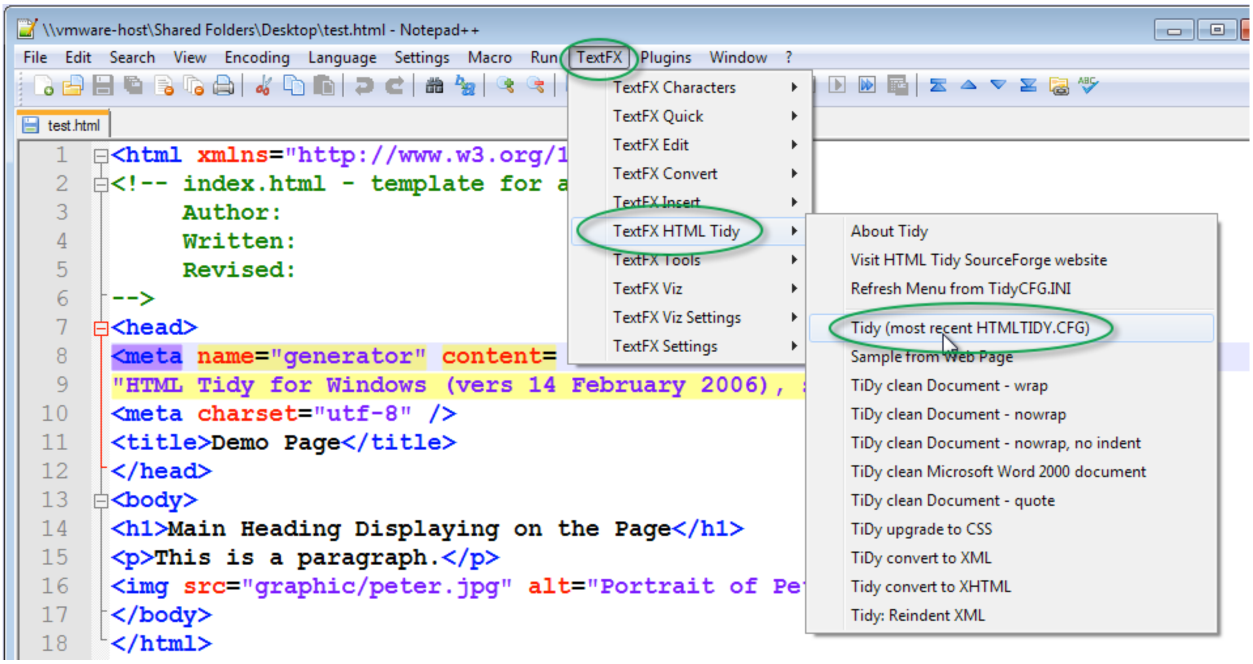
But, there are also several software programs and web sites available that will help you check your code.

The W3C (World Wide Web Consortium) has a site that will check your HTML5 code. You can either upload the file or validate a page out on the web. http://validator.w3.org/nu/

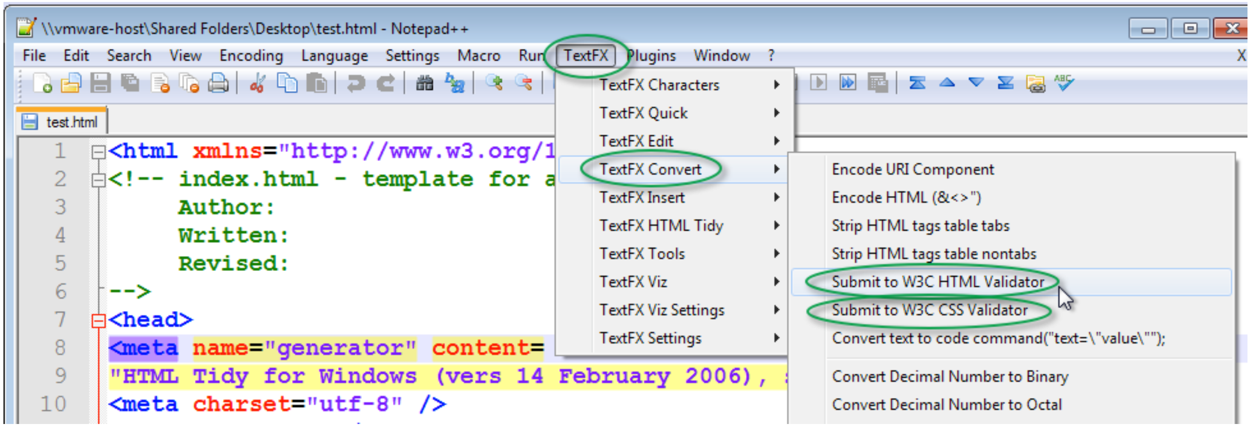


Notepad++ has a plug-in that you can download to check your code inside the text editor as part of the TextFX plugin. From the menu select TextFX/TextFX HTML Tidy/Tidy.

A new file will be opened showing the results of the validation.



You can also access the W3C Validator using TextFX in NotePad++.
From the menu item select TextFX/TextFX Convert/Submit to W3C HTML Validator.
Notice that you can also validate your CSS code by choosing the "Submit to W3C CSS Validator" as well.


\\vmware-host\Shared Folders\Desktop\test.html - Notepad++
File Edit Search View Encoding Language Settings Macro Run TextFX Plugins Window ?
test.html
<html xmlns="http://www.w3.org/1
<!-- index.html - template for a
Author:
Written:
Revised:
-->
<head>
<meta name="generator" content=
"HTML Tidy for Windows (vers 14 February 2006),
<meta charset="utf-8" />
TextFX Characters
TextFX Quick
TextFX Edit
TextFX Convert
TextFX Insert
TextFX HTML Tidy
TextFX Tools
TextFX Viz
TextFX Viz Settings
TextFX Settings
Encode URI Component
Encode HTML (&<>")
Strip HTML tags table tabs
Strip HTML tags table nontabs
Submit to W3C HTML Validator
Submit to W3C CSS Validator
Convert text to code command("text=\"value\"");
Convert Decimal Number to Binary
Convert Decimal Number to Octal

Summary

What have you learned?

You've met the father of HTML, namely SGML, as well as her sibling XHTML.

And, you've learned the five rules for valid HTML code:

1. Only one root element. That would be <html> </html>
2. Tags are always written in lower case and are case sensitive.
3. All about attributes and that the value of an attribute has to be inside of quotes.
4. That each start tag needs a stop tag; even the lonely single tag elements such as <meta /> and <img />
5. Elements must nest properly.

You also found out out the Validation tool that W3C has available so you can automatically check your code as well as the Tidy tool that comes with NotePad++.

Why Write Valid HTML Code?

As a web developer you never know what type browser the end user is going to be using.

The bottom line is this: if you write valid HTML code you know you web pages will display in all browsers. From the flexible (and huge) browsers found on a desktop machine to the teeny, tiny, itty-bitty browsers that comes with everyone's mobile phone.

You are also future proofing! By writing valid code you know your pages will display properly in browsers that haven't even been invented yet.

Credits

Peter K. Johnson - http://WebExplorations.com

July 2013